

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			5 cm ³ must give reason to keep total volume constant / to ensure concentration of hydrogensulfate(IV) stays constant			1	1		1
	(b)			first order (1) must attempt to show reasoning award (1) for suitable explanation/reasoning e.g. • when concentration (of iodate) is doubled the time taken halves • time taken is inversely proportional to concentration (of iodate) • rate goes from 0.006 to 0.012 when concentration (of iodate) doubles • rate doubles when concentration (of iodate) doubles		1	1	2	2	2
	(c)			award (1) for suitable rate equation with overall order of four e.g. rate = $k [\text{IO}_3^-] [\text{HSO}_3^-]^3$			2	2		
	(d)			award (1) for any balanced equation with $\text{IO}_3^- / \text{HSO}_3^-$ as reactants e.g. $\text{IO}_3^- + 3\text{HSO}_3^- \rightarrow \text{I}^- + 3\text{HSO}_4^-$ ecf possible from part (c)			1	1		

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	(e)	(i)		bright/canary yellow	1			1		1
		(ii)		moles of gas at 200°C = 5.653×10^{-3} mol (1) moles of gas at 20°C = 4.562×10^{-3} mol (1) [if the same error made in both calculations above then penalise once only] moles of iodine = 1.091×10^{-3} mol ⇒ percentage of iodine = 19.3% (1)			3	3	3	
				Question 7 total	1	1	8	10	5	4